EMNLP 2026

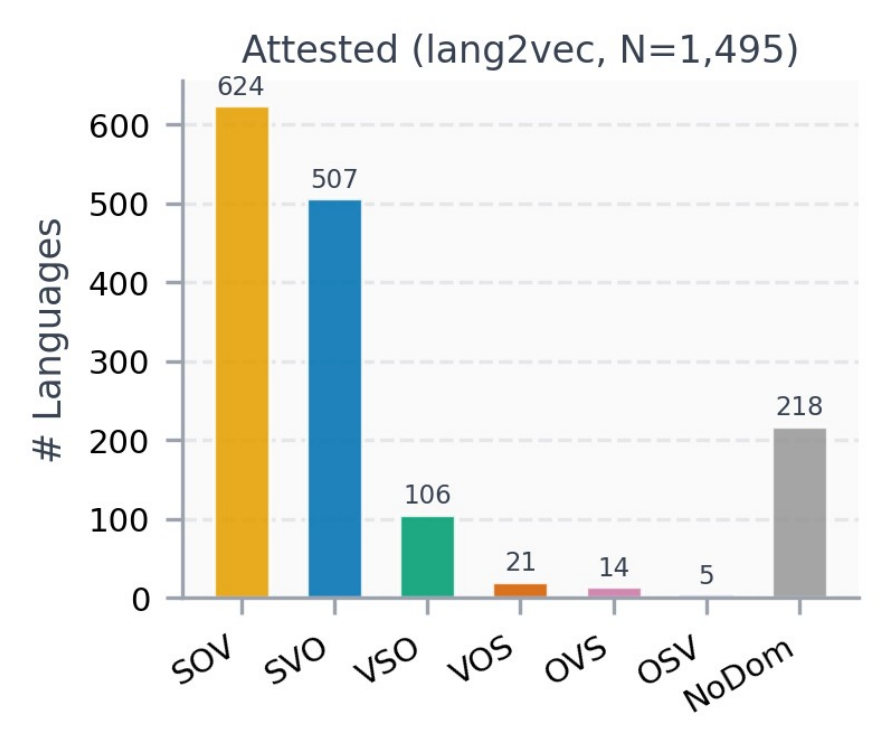
# Left-Branching Transformers Excel at Right-Branching Languages

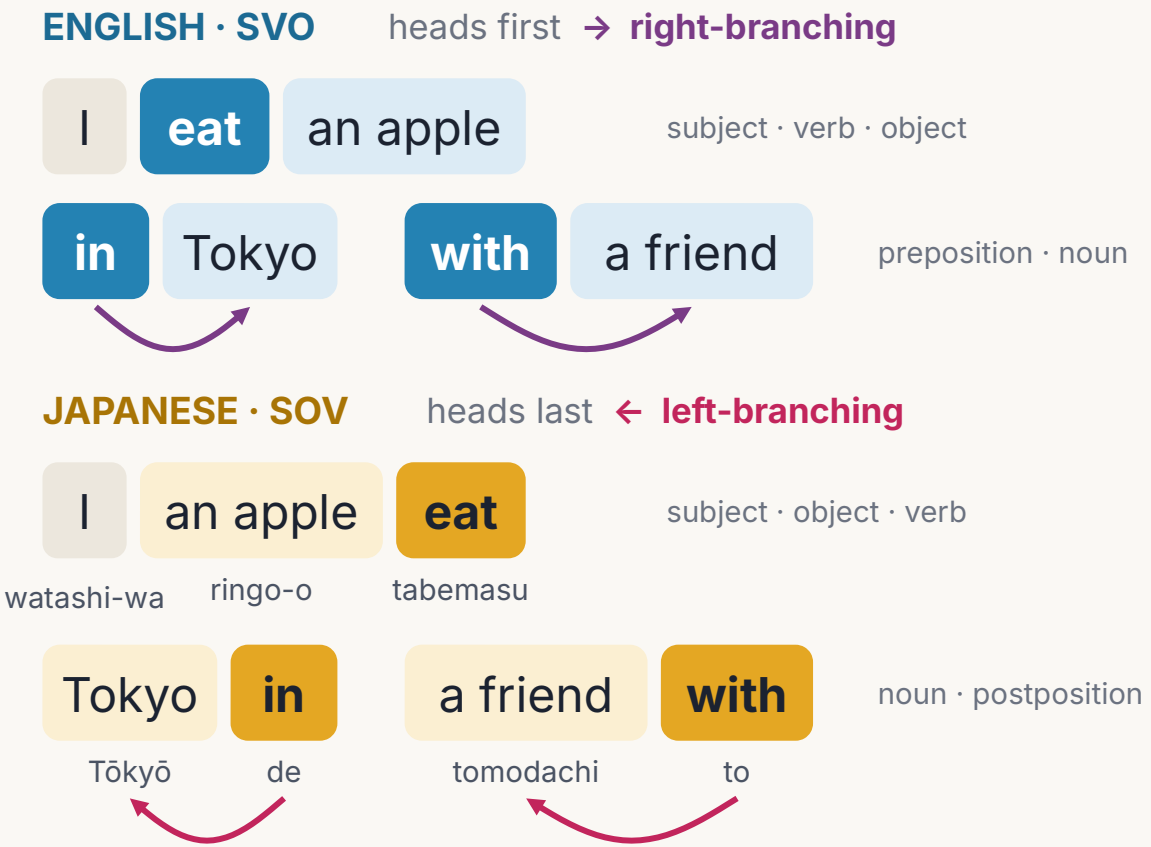
Data Shapes Word Order Preferences in Language Models

**Varvara Arzt · Allan Hanbury · Terra Blevins**

TU Wien · Northeastern University

# Natural Languages Vary a Lot in their Word Order



# Do LMs Handle All Word Orders Equally Well?

RESEARCH QUESTIONS

**RQ1** Do LMs have systematic word order preferences?

**RQ2** If there are systematic preferences, are these architectural or data-driven?

**RQ3** Do LM preferences mirror attested patterns?

**RQ4** Do LM preferences on artificial languages match those on natural languages?

**Prior work:** mixed results, partly because each study looks at only **one setting**

White & Cotterell 2021 · Kallini et al. 2024 · Kuribayashi et al. 2024 · El-Naggar et al. 2025 · Ziv et al. 2026

**Artificial languages only**

Controlled, but no semantics, morphology or discourse: findings may not transfer

**Natural languages only**

Realistic, but word order is confounded with everything else

**→ We do both: same architecture, matched training data scales**

# Experimental Setup

| | Natural languages | Artificial languages |
|---|---|---|
| **Languages** | >70 typologically diverse languages<br>Same language set at every data size | 192 PCFG-generated languages<br>6 base orders × $2^5$ internal constituent orderings |
| **Models** | Monolingual **Goldfish** (GPT-2 architecture)<br>+ multilingual mGPT, XGLM, BLOOM | **GPT-2 small**, trained from scratch<br>Matched to Goldfish at 5 & 10 MB |
| **Training data** | Web text: word order + everything else<br>5 MB → 1 GB per language, byte-premium scaled | Identical data, only word order differs<br>5 & 10 MB · shared word-boundary BPE tokeniser |
| **Evaluation** | FLORES-200 & PUD<br>BPEC (bits per English character) | Held-out sentences<br>Perplexity |

$$\mathrm{BPEC}_\ell = \frac{-\sum_{s \in S_\ell} \sum_{t=1}^{n_s} \log_2 p_\theta\left(x_t^{(s)} \mid x_{<t}^{(s)}\right)}{C_{\mathrm{eng}}}$$

**BPEC**: bits per English character

NLL ÷ English character count: comparable across tokenisers

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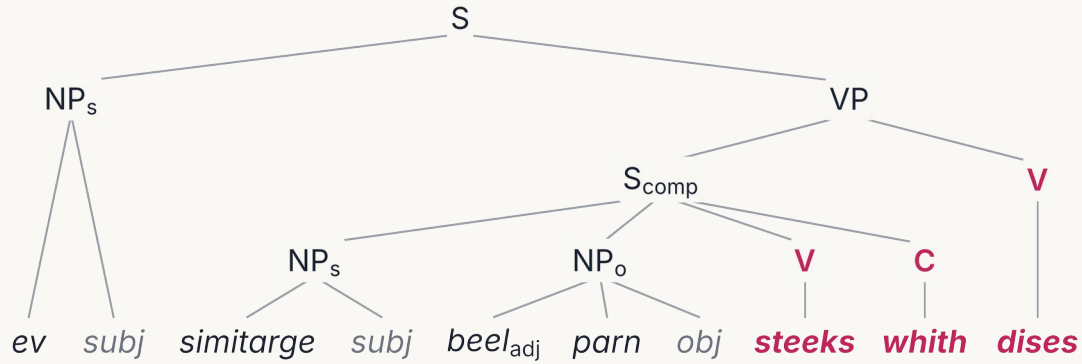
Shared across both settings: **GPT-2 architecture · 5 & 10 MB of training data**

# Building Artificial languages

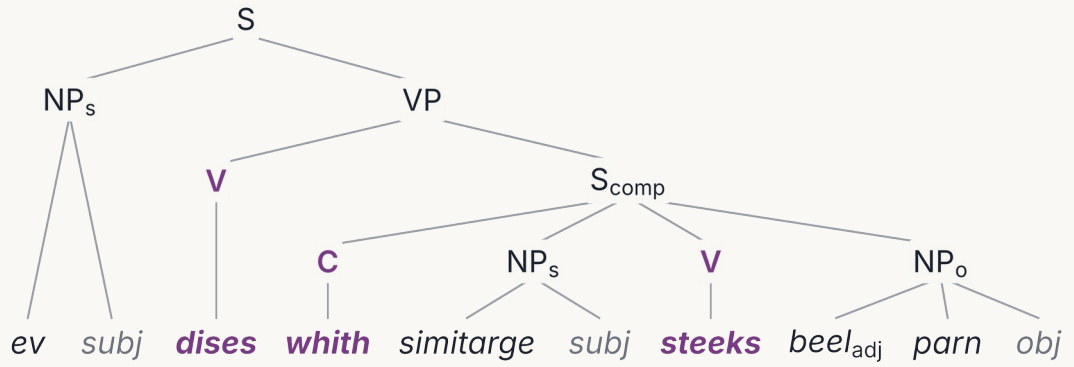
| Switch | L | R |
|---|---|---|
| 1 **Complement clause** | S-comp V | V S-comp |
| 2 **Complementiser** | S Comp | Comp S |
| 3 **Adposition** | NP Postp | Prep NP |
| 4 **Adjective** | Adj N | N Adj |
| 5 **Relative clause** | VP Rel N | N Rel VP |

**6 base orders × $2^5$ = 192 languages**

**SOV-LLLLL ~ Japanese**




**SVO-RRRLR ~ English**



# Labeling Natural Languages the Same Way

ONE LABEL PER LANGUAGE

**SVO-** **R** **R** **R** **L** **R**

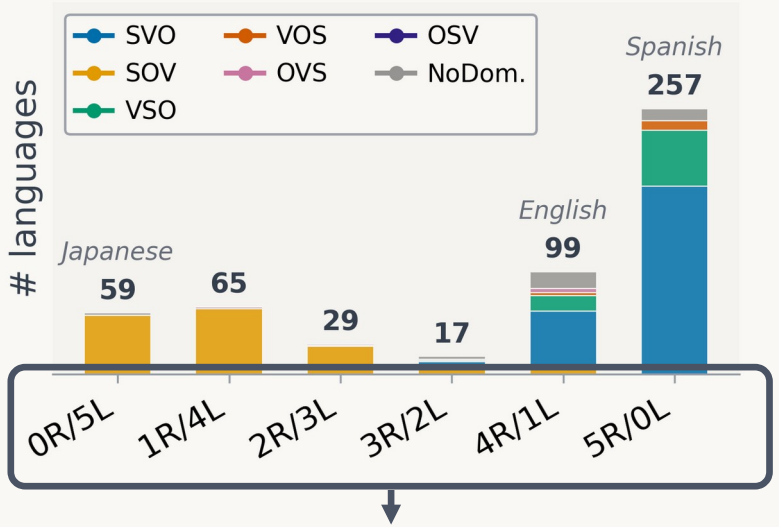
**1** complement clause | **2** comple-mentiser | **3** adposition | **4** adjective | **5** relative clause

base word order + 5 binary switches: **L** left- / **R** right-branching

| Language | Label | Scale |
|---|---|---|
| **English** | SVO-**RRRLR** | 4R/1L |
| **French** | SVO-**RRRRR** | 5R/0L |
| **Hindi** | SOV-**LRLLL** | 1R/4L |
| **Japanese** | SOV-**LLLLL** | 0R/5L |

Labels from WALS, Grambank and APiCS; only languages with all five switches annotated

**Documented natural language distribution (N=526)**




**Scale** = number of right-branching switches
**0R/5L** fully left-branching · **5R/0L** fully right-branching

# Labeling Natural Languages the Same Way

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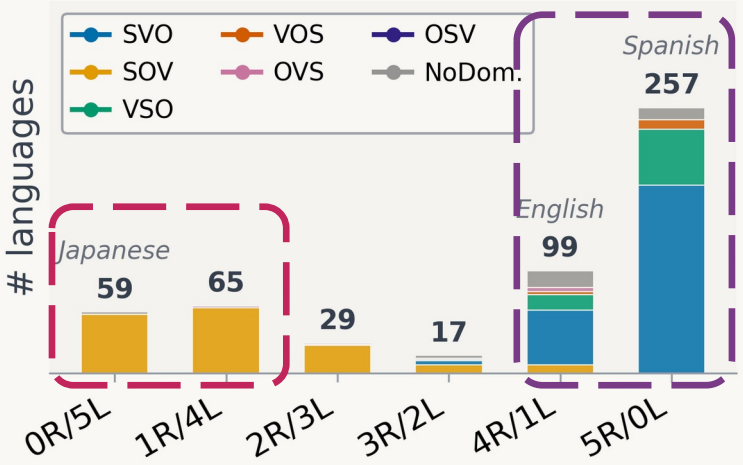
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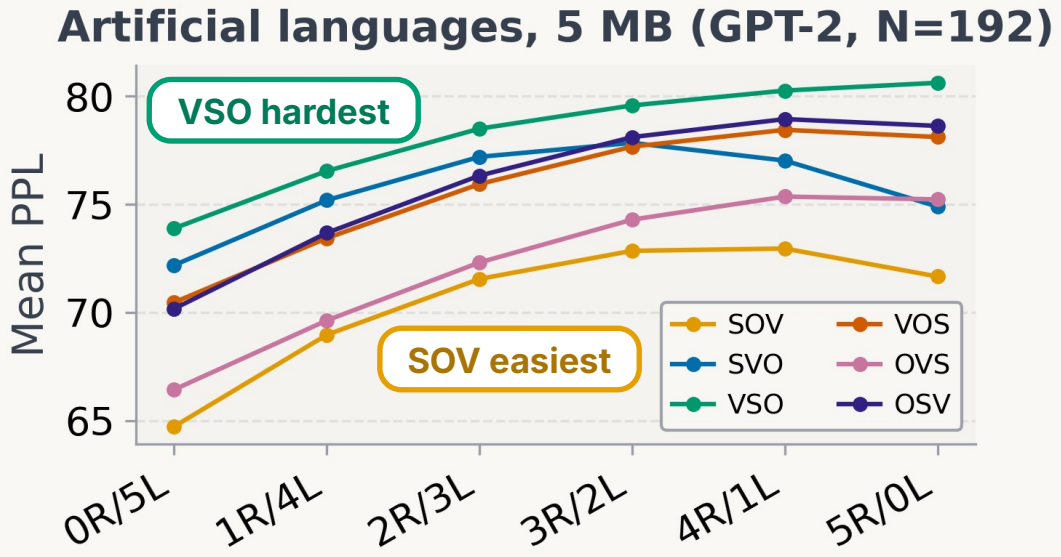

**Scale** = number of right-branching switches
**0R/5L** fully left-branching · **5R/0L** fully right-branching

Natural languages cluster at both extremes:
**mostly left-** and **mostly right-branching**

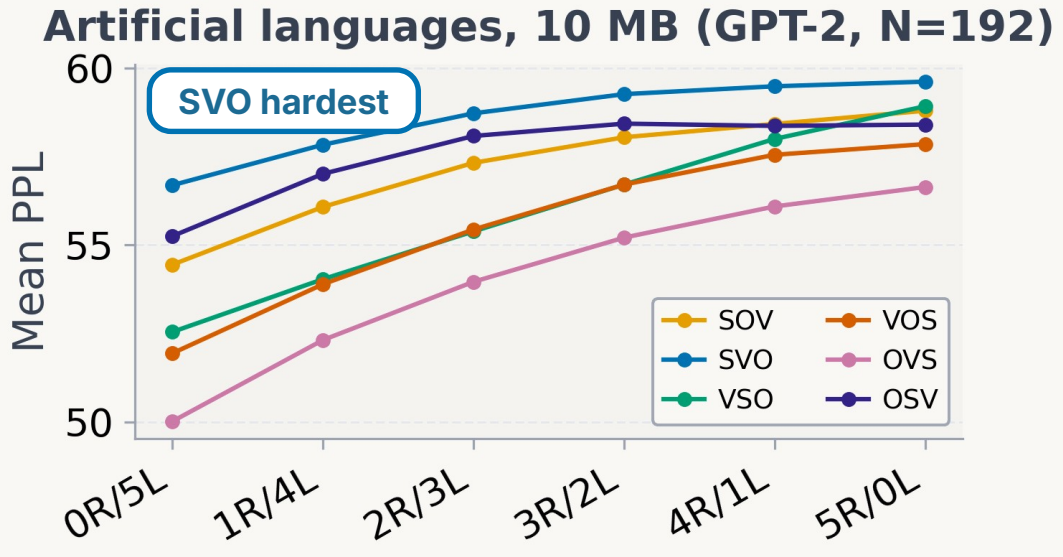
# LM Preferences in Artificial Languages

## Base word order rankings flip with more training data

GPT-2 · 5 MB




GPT-2 · 10 MB



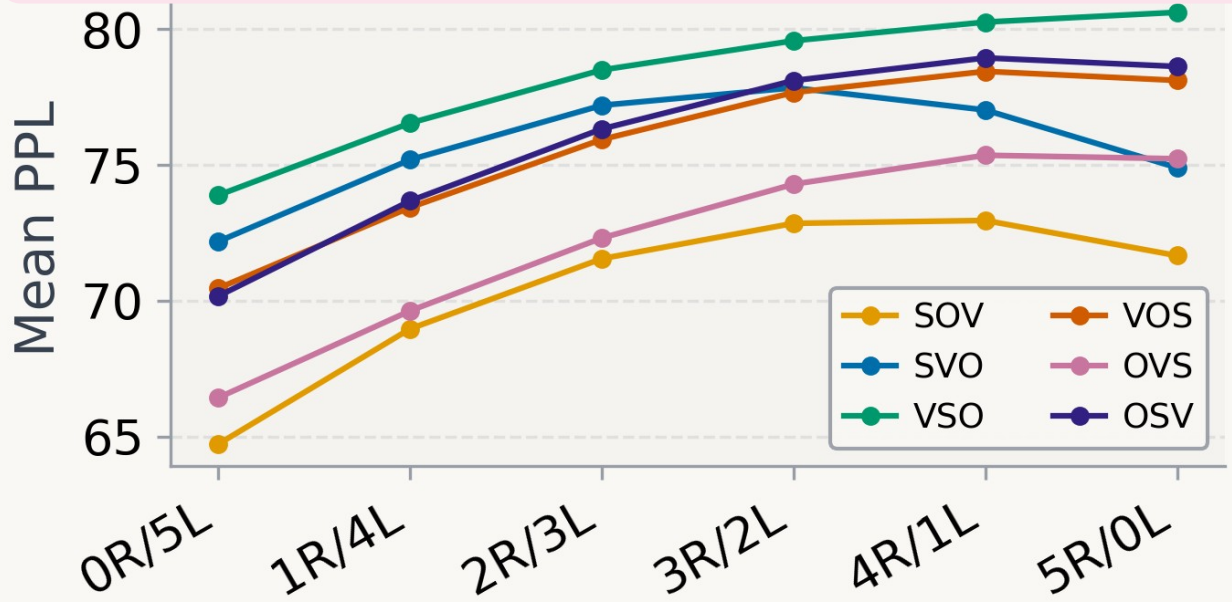

**5 MB:** **SOV** easiest, **VSO** hardest → **10 MB:** VS orders easiest, **SVO** **hardest**

No stable base word order preference; at scale on natural languages, **SVO** will be the **easiest**

# LM Preferences in Artificial Languages

## Left-branching is easiest for all six base word orders, consistently at 5 and 10 MB

**Left-branching preference also reported in prior work**
White & Cotterell, 2021; Kuribayashi et al., 2024; El-Naggar et al., 2025

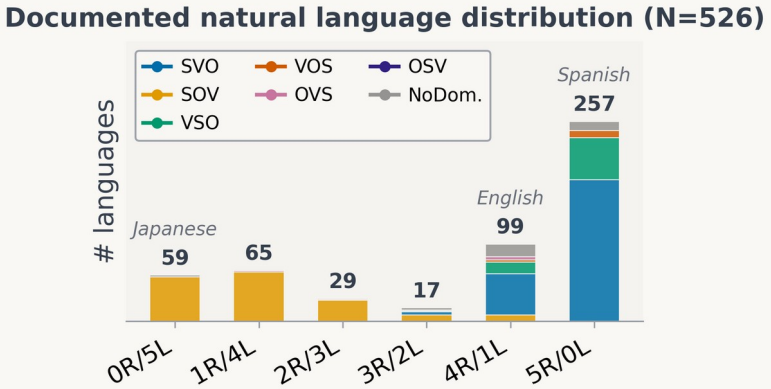



**← fully left-branching**
e.g. SOV-LLLLL ≈ Japanese

**fully right-branching →**
e.g. SVO-RRRRR ≈ French

Mean PPL over 10 runs · lower = easier

**Natural languages: SOV mostly left-, SVO mostly right-branching**

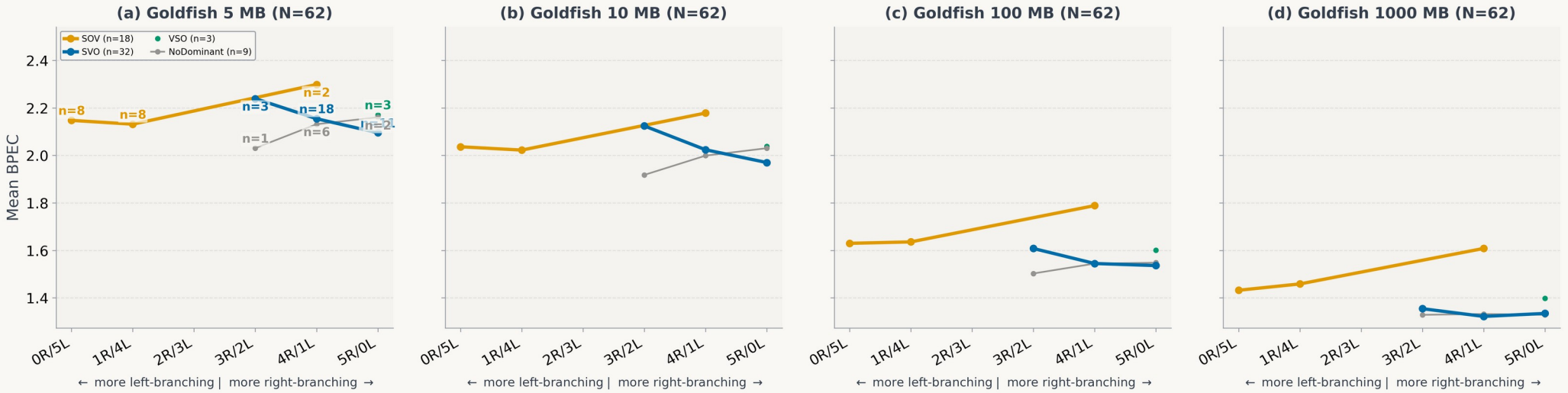



**LMs show no such pattern** on artificial languages: left-branching is easiest for every base word order

The easiest configuration for each base word order has **no natural counterpart**

# LM Preferences in Natural Languages

**No preference at 5 MB.** With more data, right-branching SVO becomes **easiest**.




| SVO–SOV δ | 0.00 | −0.10 | **−0.60*** ** | **−0.76*** ** |
|---|---|---|---|---|

**Opposite of artificial languages:** there, SVO is the **hardest** base word order at 10 MB; here, the **easiest** at scale

**Same languages at every size:** by 1 GB, SOV is ≈10% harder to model, though it is the most frequent order worldwide

**Multilingual models** (XGLM, mGPT) also find **SVO** easier than **SOV** · details in the paper

Mean BPEC on FLORES-200, lower = easier · δ = Cliff's δ, < 0: SVO easier · *** p < .001

# What Drives the SVO Advantage?

## Linear mixed-effects model: the growing SVO–SOV gap survives every control

**Not script.** 71% of SOV vs 19% of SVO FLORES-200 languages are non-alphabetic, but on alphabetic scripts only the gap still grows with data

**Not morphological complexity.** Its effect shrinks with data, so it cannot explain a gap that grows

**Measured as subword MATTR** (moving-average type–token ratio)

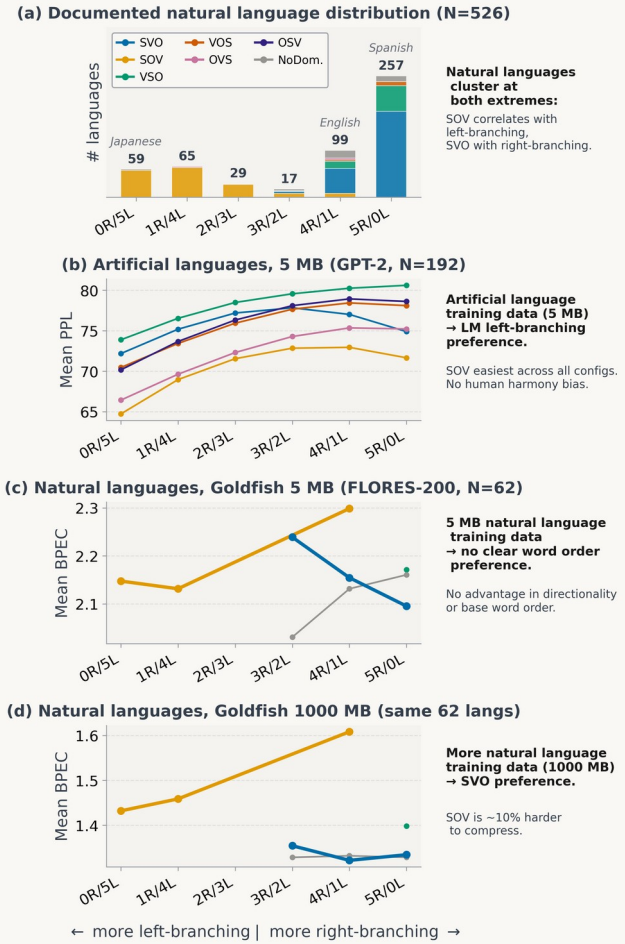
**Not training data composition.** Web-crawl (OSCAR) vs. curated share of each language's data: no effect on the growth

**Partly language resourcedness** (≈20% of the effect): a proxy for data quality

**Measured as Joshi et al. (2020) class 0–5**: how much NLP data a language has

Linear mixed-effects model, 50 SVO/SOV languages × 4 sizes: BPEC ~ word order × size + covariates + (1 | language)

# Takeaways




1 **Systematic, but not human-like: left-branching** on **artificial languages** regardless of base word order and against word order correlations in **natural languages**

2 **Data overrides the architectural bias:** on **natural languages** at scale, better-resourced, **right-branching SVO** languages become easiest, in mono- and multilingual models

3 **LM preferences on artificial and natural languages go in opposite directions at scale: SVO** is hardest on **artificial languages**, easiest on **natural ones**

**Even monolingual models trained on just 1 GB** develop the **SVO** preference; frontier models, with far larger resource gaps between languages, likely amplify it.

A precondition for ***silent typological homogenisation***: flexible word order languages drifting towards the dominant ones.

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